

Engineering Mathematics: Homogeneous Functions

A function is said to be homogeneous if scaling all its independent variables by a constant factor results in scaling the overall function value by a proportional power of that factor. Identifying homogeneous functions and determining their degree is a critical prerequisite for applying **Euler's Theorem** in partial differentiation.

Core Concept: What is a Homogeneous Function?

A function $f(x, y, z)$ is called a **homogeneous function of degree n** if, when replacing x with xt , y with yt , and z with zt (where t is a non-zero scaling factor), it satisfies the algebraic identity:

$$f(xt, yt, zt) = t^n f(x, y, z)$$

Here, n represents the **degree** of homogeneity. The degree can be any real number: positive, negative, zero, or a fraction.

Example 1: Basic Algebraic Homogeneous Function

Problem Statement

Determine if the function $u = x^2 + y^2 + z^2$ is homogeneous, and find its degree.

Step-by-Step Verification

Define the function algebraically:

$$f(x, y, z) = x^2 + y^2 + z^2$$

Substitute $x \rightarrow xt$, $y \rightarrow yt$, and $z \rightarrow zt$ into the function:

$$f(xt, yt, zt) = (xt)^2 + (yt)^2 + (zt)^2$$

Expand the individual squared terms:

$$f(xt, yt, zt) = x^2t^2 + y^2t^2 + z^2t^2$$

Factor out the common term t^2 from the entire expression:

$$f(xt, yt, zt) = t^2(x^2 + y^2 + z^2)$$

Substitute back the original function identity $u = f(x, y, z)$:

$$f(xt, yt, zt) = t^2u = t^2f(x, y, z)$$

Conclusion

Because the function cleanly factored into the form $t^2f(x, y, z)$, the function u is a **homogeneous function of degree 2**.

Example 2: Logarithmic/Fractional Composition (Non-Homogeneous Overall)

Problem Statement

Analyze the function $u = \log\left(\frac{x^3+y^3}{x^2+y^2}\right)$ to determine if it satisfies the structural conditions of homogeneity.

Step-by-Step Analysis

Define the inner algebraic rational expression as a distinct function:

$$f(x, y) = \frac{x^3 + y^3}{x^2 + y^2}$$

Substitute $x \rightarrow xt$ and $y \rightarrow yt$ into this inner algebraic component:

$$f(xt, yt) = \frac{(xt)^3 + (yt)^3}{(xt)^2 + (yt)^2}$$

Expand the exponential powers across both numerator and denominator terms:

$$f(xt, yt) = \frac{x^3t^3 + y^3t^3}{x^2t^2 + y^2t^2}$$

Factor out t^3 in the numerator and t^2 in the denominator:

$$f(xt, yt) = \frac{t^3(x^3 + y^3)}{t^2(x^2 + y^2)}$$

Simplify the powers of t using the exponent law $\frac{a^m}{a^n} = a^{m-n}$:

$$\frac{t^3}{t^2} = t^{3-2} = t^1$$

Rewrite the expression to isolate the parameter t :

$$f(xt, yt) = t^1 \cdot \left(\frac{x^3 + y^3}{x^2 + y^2} \right) = t^1 f(x, y)$$

Thus, the *inner algebraic argument* is homogeneous of degree 1. Now, substitute this result back into the full definition of the logarithmic expression u :

$$u(xt, yt) = \log \left(t^1 \cdot \left(\frac{x^3 + y^3}{x^2 + y^2} \right) \right)$$

Conclusion

Because of the outer logarithmic wrapper, you cannot factor t completely outside the function to achieve the form $t^n u$. Therefore, **the overall function u is not homogeneous**.

Crucial Transition for Euler's Theorem: While u itself is not homogeneous, the inner component is. In upcoming Euler's Theorem applications, you will learn to drop the outer logarithm by transforming the equation to $e^u = \frac{x^3 + y^3}{x^2 + y^2}$, creating a new substitute function that *is* perfectly homogeneous of degree 1.